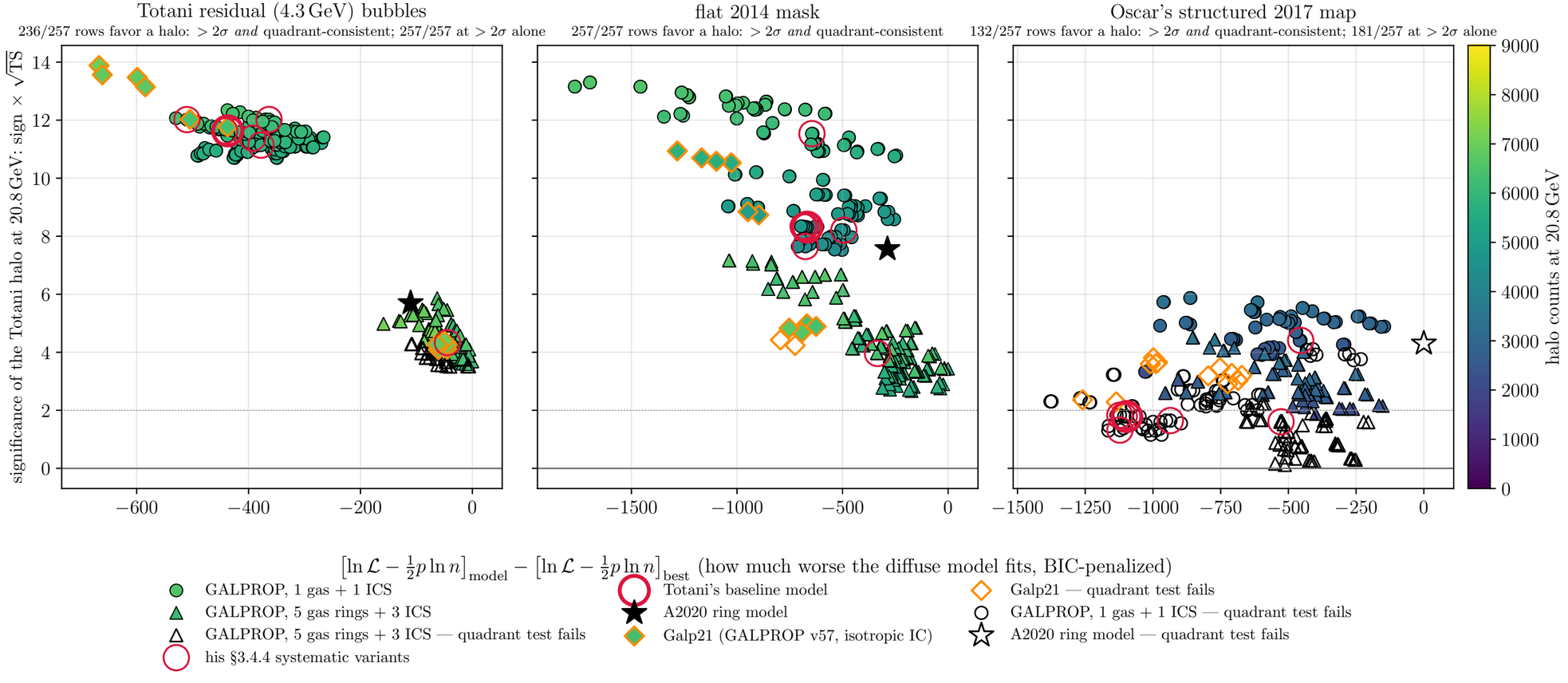

Totani residual (4.3 GeV) bubbles
236/257 rows favor a halo: > 2σ and quadrant-consistent; 257/257 at > 2σ alone
flat 2014 mask
257/257 rows favor a halo: > 2σ and quadrant-consistent
Oscar's structured 2017 map
132/257 rows favor a halo: > 2σ and quadrant-consistent; 181/257 at > 2σ alone
significance of the Totani halo at 20.8 GeV: sign × √TS
halo counts at 20.8 GeV
$\left[\ln\mathcal{L} - \frac{1}{2}p\ln n\right]_{\rm model} - \left[\ln\mathcal{L} - \frac{1}{2}p\ln n\right]_{\rm best}$ (how much worse the diffuse model fits, BIC-penalized)
GALPROP, 1 gas + 1 ICS
GALPROP, 5 gas rings + 3 ICS
GALPROP, 5 gas rings + 3 ICS — quadrant test fails
his §3.4.4 systematic variants
Totani's baseline model
A2020 ring model
Galp21 (GALPROP v57, isotropic IC)
Galp21 — quadrant test fails
GALPROP, 1 gas + 1 ICS — quadrant test fails
A2020 ring model — quadrant test fails